

What is BCD?

- BCD represents each decimal digit using 4-bit binary.
- Example:
- Decimal 5 = BCD 0101

8421 BCD Code

- Weights: 8,4,2,1
- Decimal 7 = 0111
- Decimal 9 = 1001

Decimal to BCD

- Decimal 59
- 5 = 0101
- 9 = 1001
- BCD = 0101 1001

BCD to Decimal

- BCD: 0011 0100
- 3 = 0011
- 4 = 0100
- Decimal = 34

BCD Code

Binary Coded Decimal (BCD)

- The **BCD** equivalent of a decimal number is written by **replacing each decimal digit** in the integer and fractional parts **with its four-bit binary equivalent**.
- The **BCD** code described above is more precisely known as the **8421 BCD code**.
- As an example, the BCD equivalent of **(23.15)₁₀** is written as **(0010 0011.0001 0101)_{BCD}**
- A given **BCD** number can be **converted** into an equivalent **binary** number by first **writing its decimal equivalent** and then **converting it into its binary equivalent**

Example: find the binary equivalent of the BCD number 0010 1001.0111 0101

Solution :

.BCD number: 0010 1001.0111 0101.

- Corresponding decimal number: 29.75.
- The binary equivalent of 29.75 can be determined to be 11101 for the integer part and .11 for the fractional part.
- Therefore, (0010 1001.0111 0101)BCD =(11101.11)₂.

code is the most popular of all the BCD codes, it is simply referred to as the BCD code.

- The process of binary-to-BCD conversion is the same as the process of BCD-to-binary conversion executed in reverse order. A given binary number can be converted into an equivalent BCD number by first determining its decimal equivalent and then writing the corresponding BCD equivalent.

Example: find the BCD equivalent of the binary number 10101011.101:

Solution :

- The decimal equivalent of this binary number can be determined to be 171.625.
- The BCD equivalent can then be written as 0001 0111 0001.0110 0010 0101.
- **Valid & Invalid Codes**

Valid: 0000 to 1001

Invalid: 1010 to 1111

BCD Addition

- Add normally
- If result >9 add 0110
- Example:
- $5+7 = 12$
- BCD correction needed

BCD ADDITION

Rule:

Add two BCD numbers normally (binary addition).

If:

- Result > 1001 (9) **OR**
- Carry is generated
 ➡ **Add 0110 (6)** for correction.

✓ Example 1:

$$5 + 4$$

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$$5 = 0101$$

$$4 = 0100$$

$$1001 \quad (9) \rightarrow \text{VALID}$$

✓ No correction needed

Answer = 9

✓ Example 2:

$$7 + 6$$

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$$7 = 0111$$

$$6 = 0110$$

$$1101 \quad (13) \rightarrow \text{INVALID}$$

Add correction **0110**

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$$1101$$

$$+0110$$

$$1 \ 0011$$

Carry = 1

Result = 13

✓ Example 3:

$$9 + 5$$

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$$9 = 1001$$

$$5 = 0101$$

$$1110 \quad (14) \rightarrow \text{INVALID}$$

Add 0110

markdown

$$1110$$

$$+0110$$

$$1 \ 0100$$

Answer = 14

◆ BCD SUBTRACTION

Method:

Use 10's complement

Steps:

1. Take 9's complement
2. Add 1 → 10's complement
3. Add to minuend
4. If carry → positive result

If no carry → take 10's complement again & add minus sign



Example 1:

$$45 - 23$$

Step 1: 9's complement of 23

$$9 - 2 = 7$$

$$9 - 3 = 6$$

$$\rightarrow 76$$

Step 2: +1

$$76 + 1 = 77$$

Step 3: Add to 45

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45

+77

122

Ignore carry $\rightarrow 22$



Correct Answer = 22



Example 2:

$$32 - 48$$

9's complement of 48

51

$$+1 \rightarrow 52$$

Add:

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32

+52

84

No carry $\rightarrow$ Negative result

Take 10's complement of 84

diff

$$9's \text{ comp} = 15$$

$$+1 = 16$$



Final Answer = -16

◆ Quick Summary

Operation	Rule
BCD Addition	If $>9 \rightarrow$ add 0110
BCD Subtraction	Use 10's complement

Practice for Students

1. $6 + 9$
2. $8 + 7$
3. $54 - 29$
4. $37 - 58$

Advantages

- Easy display
- Human readable
- Used in calculators

Disadvantages

- Needs more bits
- Slower than binary

Applications

- Digital clocks
- Calculators
- Seven segment displays

Quiz

- 1) Write BCD for 45
- 2) Convert 1001 0010 to decimal
- 3) Why correction needed?